

Translated

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15 Puzzle Game: Existence Of The Solution

This game is played on a 4×4 board. On this board there are 15 playing tiles numbered from 1 to 15. One cell is left empty (denoted by 0). You need to get the board to the position presented below by repeatedly moving one of the tiles to the free space:

$$\begin{matrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \\ 9 & 10 & 11 & 12 \\ 13 & 14 & 15 & 0 \end{matrix}$$

The game "15 Puzzle" was created by Noyes Chapman in 1880.

Existence Of The Solution

Let's consider this problem: given a position on the board, determine whether a sequence of moves which leads to a solution exists.

Suppose we have some position on the board:

$$\begin{matrix} a_1 & a_2 & a_3 & a_4 \\ a_5 & a_6 & a_7 & a_8 \\ a_9 & a_{10} & a_{11} & a_{12} \\ a_{13} & a_{14} & a_{15} & a_{16} \end{matrix}$$

where one of the elements equals zero and indicates an empty cell $a_z = 0$

Let's consider the permutation:

$$a_1 a_2 \dots a_{z-1} a_{z+1} \dots a_{15} a_{16}$$

i.e. the permutation of numbers corresponding to the position on the board without a zero element

Let N be the number of inversions in this permutation (i.e. the number of such elements a_i and a_j that $i < j$, but $a_i > a_j$).

Suppose K is an index of a row where the empty element is located (i.e. using our convention, $K = (z - 1) \div 4 + 1$).

Then, **the solution exists iff $N + K$ is even.**